

Experiment / Case Study No.: 7

1. Experiment Title:

Perform DOM Traversal and Updates; Develop a To-Do List App with Add, Edit, and Delete Features

2. Software/Tools Required:

Visual Studio Code, HTML5, CSS3, JavaScript (DOM Traversal, Event Handling), Google Chrome Browser

3. Experiment Program Code:

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Focus Flow | Task Manager</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>
  <div class="container">
    <div class="header">
      <h1>Focus Flow</h1>
      <p>Organize your day with clarity</p>
    </div>

    <div class="input-group">
      <input type="text" id="taskInput" placeholder="Enter a new task...">
      <button id="addBtn">Add Task</button>
    </div>

    <div class="task-info">
      <span id="taskCount">0 tasks pending</span>
    </div>

    <ul id="taskList">
    </ul>

    <p id="emptyMessage" class="hidden">Start by adding your first task.</p>
  </div>

  <script src="script.js"></script>
</body>

</html>
```

script.js

```
var taskInput = document.getElementById('taskInput');
var addBtn = document.getElementById('addBtn');
var taskList = document.getElementById('taskList');
var taskCount = document.getElementById('taskCount');
var emptyMessage = document.getElementById('emptyMessage');

updateTaskCount();

addBtn.addEventListener('click', addTask);

taskInput.addEventListener('keypress', function (event) {
    if (event.key === 'Enter') {
        addTask();
    }
});

function addTask() {
    var taskText = taskInput.value.trim();

    if (taskText === '') {
        alert('Please enter a task!');
        return;
    }

    var li = document.createElement('li');

    var span = document.createElement('span');
    span.className = 'task-text';
    span.textContent = taskText;

    var editBtn = document.createElement('button');
    editBtn.textContent = 'Edit';
    editBtn.className = 'edit-btn';

    var deleteBtn = document.createElement('button');
    deleteBtn.textContent = 'Delete';
    deleteBtn.className = 'delete-btn';

    li.appendChild(span);
    li.appendChild(editBtn);
    li.appendChild(deleteBtn);

    taskList.appendChild(li);

    taskInput.value = '';
    taskInput.focus();

    editBtn.addEventListener('click', function () {
        var currentText = span.textContent;
        var newText = prompt('Edit your task:', currentText);

        if (newText !== null && newText.trim() !== '') {
            span.textContent = newText.trim();
        }
    });
}
```

```

    });

    deleteBtn.addEventListener('click', function () {
        var parent = this.parentElement;
        parent.remove();
        updateTaskCount();
    });

    updateTaskCount();
}

function updateTaskCount() {
    var tasks = taskList.children;
    var count = tasks.length;

    taskCount.textContent = count + ' task' + (count !== 1 ? 's pending' : '
pending');

    if (count === 0) {
        emptyMessage.classList.remove('hidden');
    } else {
        emptyMessage.classList.add('hidden');
    }
}

```

4. Output:

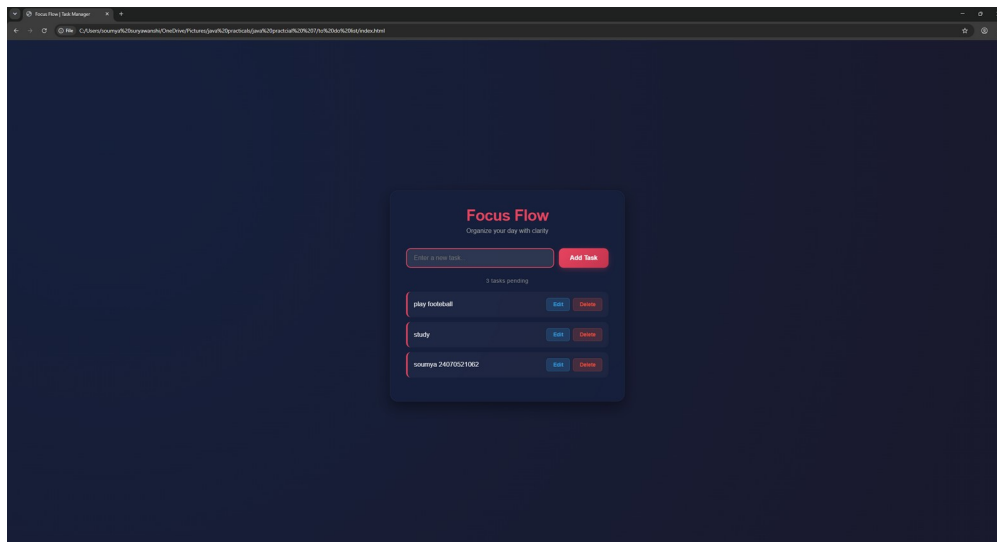


Fig 4: Focus Flow To-Do List output — PRN: 24070521062, Name: Soumya Suryawanshi, File Path: C:/Users/soumya suryawanshi/OneDrive/Pictures/java practicals/java practcial 207/to do list/index.html

5. Case Study Title:

Registration (Sign Up) Form — Demonstrating JavaScript Events Handling: Accessing and Manipulating Form Elements, Validating User Input, and Using Form Events (submit, change, focus)

6. Case Study Program Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Registration Form Case Study</title>
  <link rel="stylesheet" href="style.css">
  <link href="https://fonts.googleapis.com/css2?
family=Inter:wght@400;500;600&display=swap" rel="stylesheet">
</head>
<body>
  <div class="form-container">
    <h2>Sign Up</h2>
    <p class="mandatory-text"><span class="asterisk">*</span> Mandatory to
fill</p>

    <form id="signupForm" novalidate>
      <div class="form-group">
        <label for="firstname">Firstname: <span
class="asterisk">*</span></label>
        <input type="text" id="firstname" name="firstname" required
placeholder="Enter your first name">
        <span class="error-message" id="firstnameError"></span>
      </div>

      <div class="form-group">
        <label for="lastname">Lastname:</label>
        <input type="text" id="lastname" name="lastname" placeholder="Enter
your last name">
      </div>

      <div class="form-group">
        <label>Birthday:</label>
        <div class="date-group">
          <select id="day" name="day">
            <option value="">Day</option>
          </select>
          <select id="month" name="month">
            <option value="">Month</option>
          </select>
          <select id="year" name="year">
            <option value="">Year</option>
          </select>
        </div>
      </div>

      <div class="form-group">
        <label for="username">Username: <span
class="asterisk">*</span></label>
        <input type="text" id="username" name="username" required
placeholder="Choose a username">
        <span class="error-message" id="usernameError"></span>
      </div>
    </form>
  </div>
</body>
</html>
```

```

    </div>

    <div class="form-group">
      <label for="email">E-mail:</label>
      <input type="email" id="email" name="email"
placeholder="example@email.com">
      <span class="error-message" id="emailError"></span>
    </div>

    <div class="form-group">
      <label for="website">Website:</label>
      <input type="url" id="website" name="website"
placeholder="https://yourwebsite.com">
    </div>

    <div class="form-group">
      <label for="password">Password: <span
class="asterisk">*</span></label>
      <input type="password" id="password" name="password" required
placeholder="Enter password">
      <span class="error-message" id="passwordError"></span>
    </div>

    <div class="form-group">
      <label for="repassword">Re-password: <span
class="asterisk">*</span></label>
      <input type="password" id="repassword" name="repassword" required
placeholder="Confirm password">
      <span class="error-message" id="repasswordError"></span>
    </div>

    <div class="form-group checkbox-group">
      <input type="checkbox" id="terms" name="terms" required>
      <label for="terms">I agree to the terms & conditions. <span
class="asterisk">*</span></label>
      <span class="error-message" id="termsError"></span>
    </div>

    <button type="submit" id="submitBtn">Sign up</button>
  </form>
</div>

<script src="script.js"></script>
</body>
</html>

```

script.js

```

document.addEventListener('DOMContentLoaded', () => {

  const form = document.getElementById('signupForm');
  const firstnameInput = document.getElementById('firstname');
  const usernameInput = document.getElementById('username');
  const emailInput = document.getElementById('email');

```

```

const passwordInput = document.getElementById('password');
const repasswordInput = document.getElementById('repassword');
const termsCheckbox = document.getElementById('terms');

populateDates();

const allInputs = form.querySelectorAll('input, select');
allInputs.forEach(input => {
    input.addEventListener('focus', function() {
        const formGroup = this.closest('.form-group');
        if (formGroup) {
            formGroup.classList.add('focused');
        }
    });

    input.addEventListener('blur', function() {
        const formGroup = this.closest('.form-group');
        if (formGroup) {
            formGroup.classList.remove('focused');
        }

        if(this.value.trim() !== '' && this.type !== 'email') {
            clearError(this);
        }
    });
});

emailInput.addEventListener('change', function() {
    if (this.value && !isValidEmail(this.value)) {
        showError(this, 'Please enter a valid email address.');
```

```

    if (passwordInput.value.trim() === '') {
        showError(passwordInput, 'Password is required');
        isValid = false;
    } else {
        clearError(passwordInput);
    }

    if (repasswordInput.value.trim() === '') {
        showError(repasswordInput, 'Please confirm your password');
        isValid = false;
    } else if (passwordInput.value !== repasswordInput.value) {
        showError(repasswordInput, 'Passwords do not match');
        isValid = false;
    } else {
        clearError(repasswordInput);
    }

    if (!termsCheckbox.checked) {
        showError(termsCheckbox, 'You must agree to the terms');
        isValid = false;
    } else {
        clearError(termsCheckbox);
    }

    if (emailInput.value && !isValidEmail(emailInput.value)) {
        showError(emailInput, 'Please enter a valid email address');
        isValid = false;
    }

    if (isValid) {
        alert('Form submitted successfully!');
        form.reset();
    }
});

function showError(inputElement, message) {
    const formGroup = inputElement.closest('.form-group');
    if (formGroup) {
        formGroup.classList.add('error');
        const errorElement = formGroup.querySelector('.error-message');
        if (errorElement) {
            errorElement.innerText = message;
        }
    }
}

function clearError(inputElement) {
    const formGroup = inputElement.closest('.form-group');
    if (formGroup) {
        formGroup.classList.remove('error');
    }
}

function isValidEmail(email) {

```

```

    const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    return emailPattern.test(email);
}

function populateDates() {
    const daySelect = document.getElementById('day');
    const monthSelect = document.getElementById('month');
    const yearSelect = document.getElementById('year');

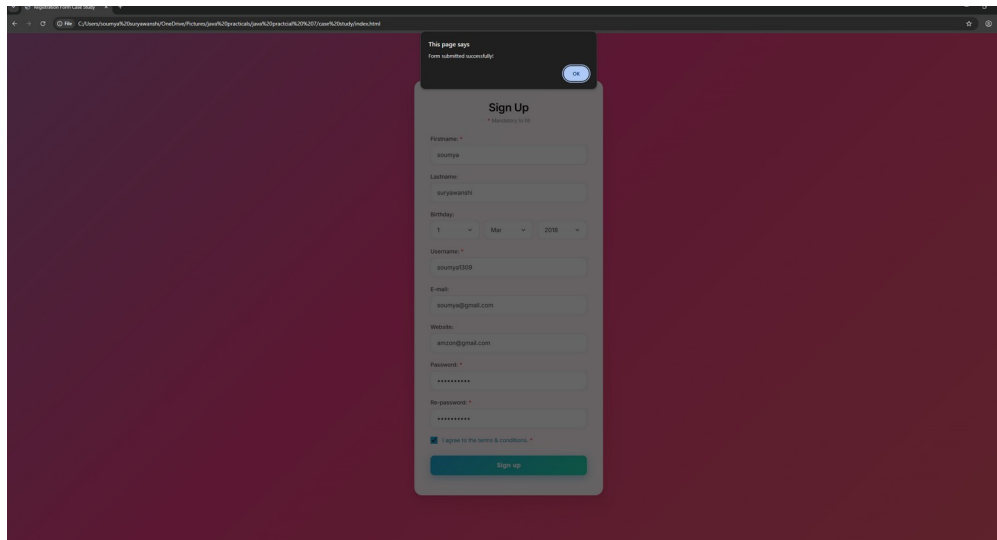
    for (let i = 1; i <= 31; i++) {
        let option = document.createElement('option');
        option.value = i;
        option.text = i;
        daySelect.appendChild(option);
    }

    const months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug",
"Sep", "Oct", "Nov", "Dec"];
    months.forEach((month, index) => {
        let option = document.createElement('option');
        option.value = index + 1;
        option.text = month;
        monthSelect.appendChild(option);
    });

    const currentYear = new Date().getFullYear();
    for (let i = currentYear; i >= 1900; i--) {
        let option = document.createElement('option');
        option.value = i;
        option.text = i;
        yearSelect.appendChild(option);
    }
}
});

```


7. Output:



The image shows a web browser window displaying a 'Sign Up' form. The form is centered on a dark red background. It includes fields for Firstname, Lastname, Birthdate, Username, E-mail, Password, and Re-password. A 'Sign up' button is at the bottom. A small notification box at the top left says 'This page says: You submitted successfully!'.

Fig 7: Sign Up Form output — PRN: 24070521062, Name: Soumya Suryawanshi, File Path: C:/Users/soumya suryawanshi/OneDrive/Pictures/java practicals/java practcial 207/case study/index.html

8. Result/Conclusion:

A To-Do List application (Focus Flow) was successfully designed and implemented using JavaScript DOM traversal and manipulation, allowing tasks to be dynamically added, edited in place using the `closest()` and `parentElement` traversal methods, deleted, and counted with a live pending-task indicator. In the accompanying case study, a Sign Up form was built to demonstrate JavaScript event handling in practice: form elements were accessed and manipulated on focus and blur events, user input was validated on submit (required fields, password match, and a regex-based email check) and on the email field's change event, and appropriate error messages were displayed inline for each field. Both implementations produced the expected output for the given inputs and reinforced practical understanding of DOM traversal, dynamic UI updates, and form-event-driven validation in JavaScript.

Upload the Soft copy as well as complete code on GitHub link is mandatory.